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Mini project 1

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5. Mini project 1

Detection of toothbrush bristle defects is carried out mainly using vision systems and machine learning. Data for this type of task includes high-resolution images and scanning electron microscope (SEM) photographs, which makes it possible to analyse the quality of bristle tips.

In production systems (quality control), the most commonly analysed are:

- **Splayed/spread bristles** — deformation of the bristles.
- **Abrasion** — damage to the bristle tips.
- **Errors in fusing/mounting** (*bristle stapling defects*) — incorrect seating of tufts.
- **Improper rounding of tips.**
- **Missing bristles.**

5.1 Project objective

The objective of the project is to develop and test the concept of a vision system for automatic detection of selected toothbrush bristle defects. As part of the project, it is necessary to:

- prepare the data (images) and define defect classes,
- propose and implement a preprocessing stage (e.g. normalisation, denoising, contrast enhancement),
- determine the regions relevant for analysis (e.g. segmentation of bristle tips) or features describing the defects,
- select and train a classification/detection model (or build a rule-based detector using the features),
- evaluate the quality of the solution (e.g. accuracy, precision/recall, confusion matrix) and discuss its limitations.

The dataset is available here: <https://drive.google.com/drive/folders/1nXdqY60uZIEWWwLzuxtZrkXygI2xNbS6?usp=sharing>.

5.2 Example literature

- Bao, Nengsheng, et al. *A deep learning-based process monitoring system for tooth-brush manufacturing defect characterization*. *Procedia CIRP* **118** (2023): 1072–1077.
- Bergmann, Paul, et al. *MVTec AD—A comprehensive real-world dataset for unsupervised anomaly detection*. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2019).